Magnetic separation (8 points)

Magnetic separation is used to separate materials of different magnetic properties. Deposits of titanium ores, for example, contain minerals such as zircon ($ZrSiO_4$), monazite ($[Ce, La, Th]PO_4$), ilmenite ($FeTiO_3$) and rutile (TiO_2), among which the titanium-bearing minerals are of great economic value. Rutile and ilmenite are used to make non-toxic pigments for inks, plastics, ceramics and much more, so they must be separated from titanium ores. Magnetic separation uses magnetic forces to separate magnetic materials. The principle of a magnetic separator is shown in Fig. 1.

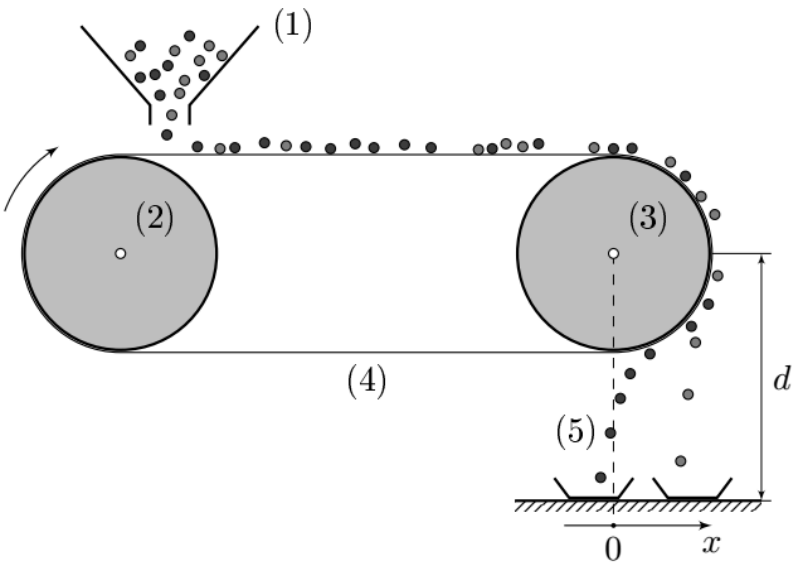


Figure 1. (1) hopper; (2) non-magnetic roller L_1; (3) magnetic roller L_2; (4) conveyor belt; (5) magnetic particles; (6) collecting trays.

The main part of the separator consists of two rollers L_1 and L_2 (diameter D) connected by a conveyor belt (Fig. 1). The thickness of the belt is negligible compared with the radius of the rollers. Roller L_1 is non-magnetic. Roller L_2 is magnetic, and the magnetic field around it varies as

$$B(r) = |\vec{B}| = B_0 e^{-r/a}, \tag{1}$$

where r is the distance from the surface of the roller in the radial direction and a is a constant. The magnetic roller rotates with angular velocity ω kept by an electric motor. Mineral particles fall from the hopper onto the belt and are carried to the magnetic roller. Friction between the belt and the magnetic roller is large enough to prevent slipping.

The following formulas may be useful:

$$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \begin{cases} 1 + e^{-2x}, & x \to \infty \\ \frac{1}{x} + \frac{x}{3} - \frac{x^3}{45}, & x \to 0 \end{cases}, \qquad e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \approx 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

In this problem we discuss a model of the separator taking into account only magnetic and gravitational forces.

Part A. Magnetic susceptibility of materials

We study the properties of paramagnetic materials. Consider a mineral containing magnetic atoms X at a concentration n, each with a magnetic moment p. The concentration is low, so the interaction between the atomic moments is negligible. Without an external field the moments point randomly and the magnetization $\vec{M}$ (the magnetic moment per unit volume) is zero. In an external field $\vec{B}$ the atoms align along the field and the material acquires a non-zero magnetization.

According to quantum mechanics the projection of an atom's magnetic moment on the field direction (the z axis) is $p_z = g_j \mu_B m$, where m takes the values $\{j, j-1, \ldots, -j\}$, j is a non-negative integer or half-integer characterizing the atom X, $\mu_B = 0.93 \cdot 10^{-23}\ \mathrm{A \cdot m^2}$ is the Bohr magneton and g_j is the Landé factor. The probability that the projection takes the value p_z is $W(p_z) = Ae^{-U_B/kT}$, where A is a normalization constant, U_B the energy of the moment in the field, T the temperature and k Boltzmann's constant. The mean projections on the x and y axes vanish.

A.1 Find the mean value $\langle p_z \rangle$ of the z component of the magnetic moment of atom X in a field $\vec{B}$ at temperature T, as a function of the parameter $\alpha = \frac{g_j \mu_B B}{kT}$. 1.1pt

A.2 Give $\langle p_z \rangle$ in the limit $\alpha \to 0$ (magnetic energy much smaller than thermal energy) and in the limit $\alpha \to \infty$. Sketch $\langle p_z \rangle$ as a function of α. 1.1pt

A.3 The magnetic susceptibility characterizes the magnetic properties of a material; in a weak field $\vec{B}$ it is defined as $\chi = \mu_0 M/B$, where M is the magnetization and μ_0 the vacuum permeability. Take the field to be weak, $\alpha \ll 1$. Calculate χ in terms of the concentration n, the temperature T, the Landé factor g_j, the parameter j and fundamental constants. 0.9pt

Part B. The magnetic separator

Let us now consider the model of the separator described above. The magnetic field B at a distance r from the surface of the magnetic roller along the radius is given in the table:

r, mm	B, T	r, mm	B, T	r, mm	B, T
1	0.606	5	0.098	9	0.024
2	0.415	6	0.070	10	0.014
3	0.262	7	0.045	11	0.008
4	0.144	8	0.030	12	0

B.1 Using the table, determine the numerical values of B_0 and a in eq. (1). Write the formulas you used. 0.8pt

B.2 From here on use the values $D = 0.04$ m, $R_0 = 0.5 \cdot 10^{-3}$ m, $a = 3.0 \cdot 10^{-3}$ m, $B_0 = 1.0$ T, $d = 1.0$ m, $\omega_0 = 2\pi \cdot 240\ \mathrm{min}^{-1}$, $\rho_1 = 5.0 \cdot 10^3\ \mathrm{kg/m^3}$, $\rho_2 = 4.3 \cdot 10^3\ \mathrm{kg/m^3}$, $\chi_1 = 6.3 \cdot 10^{-4}$, $\chi_2 = 6.5 \cdot 10^{-5}$, $\Delta = 0.2$ m. 0.8pt

Consider spherical particles of radius R_0 of one paramagnetic mineral with mean susceptibility χ_1 and density ρ_1; their magnetic moments in the field of the roller sit at their centres. The collecting tray is a distance d below the magnetic roller (Fig. 1). The roller turns at ω_0. Interactions between the particles' moments, and the magnetic force on particles that have left the roller, are negligible. When the field changes by $\mathrm{d}B$ the energy of a particle changes by $\mathrm{d}E = -\vec{P} \cdot \mathrm{d}\vec{B}$, where $\vec{P}$ is its magnetic moment.

What is the energy of a particle in the field $\vec{B}$? Express it in terms of B, the susceptibility χ_1 and the volume V of the particle.

B.3 Find the expression for the position x of the tray at which it collects the particles, and its numerical value. 1.1pt

B.4 Suppose the particles are spheres of different radii, the relative deviation from the mean not exceeding 10%: $|(R_0 - R)/R_0| \leq 0.1$, where R is the radius of a particle and R_0 the mean radius. The roller turns at ω_0. Find the expression and the numerical value of the width Δ_0 of the particle beam along x near the tray. 1.1pt

B.5 A mixture consists of particles of two minerals with susceptibilities χ_1, χ_2 and densities ρ_1, ρ_2; the mean radius of both is R_0. Find the expression for the angular velocity ω of the magnetic roller at which the two kinds can be collected separately, and its numerical value. The width of both trays is Δ. 1.1pt